

SUPERVISOR-STUDENT MATCHING AND NLP- BASED PROJECT GUIDANCE SYSTEM

Project ID: 25-26J-374

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology

Specializing in Information Technology

Department of Information Technology

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Declaration

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I declare that this is my own work and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text. Also, I hereby grant to Sri Lanka Institute of Information Technology, the nonexclusive right to reproduce and distribute my dissertation, in whole or in part in print, electronic or other medium. I retain the right to use this content in whole or part in future works (such as articles or books).

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The supervisors should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

m. n. s. l. R. L.

Signature of the supervisor

(Prof. Samantha Rajapaksha)

29/8/25

Date

Abstract

Effective supervision is a critical factor in ensuring the success of academic projects, yet many institutions continue to rely on manual allocation methods that are time-consuming, subjective, and prone to mismatches. To address this challenge, this project proposes the design and implementation of a supervisor–Student Matching and NLP-Based Project Guidance System. The primary objective is to automate the process of aligning students with supervisors by analysing project topics and comparing them with supervisor expertise and availability. This approach reduces administrative overhead while ensuring a fairer and more accurate allocation of supervision resources. The system employs Natural Language Processing (NLP) techniques to extract keywords, preprocess submitted topics, and generate semantic representations that reflect their intended meaning. These representations are compared with supervisor profiles to calculate similarity scores, resulting in a ranked list of the most suitable supervisors for each project. Alongside allocation, the system provides a dedicated communication channel to sustain effective interaction. A neural-network-based chatbot is integrated to support students with routine project-related queries and clarifications, thereby reducing delays in obtaining information. Supervisors remain directly engaged with students through the platform for guidance and academic guidance, while the chatbot complements this process by addressing frequently asked or general questions. The platform will be developed as a web application with a frontend built using HTML, CSS, and JavaScript, and a backend implemented in Python with Flask to manage requests, matching logic, and the chat interface. Descriptive indicators such as message frequency and response times will be tracked to enhance transparency and ensure smoother supervisory interactions. By combining intelligent allocation with structured communication support, the system is expected to improve supervisor–student alignment, enhance efficiency, and contribute to more effective project management in academic environments. **Keywords: Academic project management, expertise alignment, guidance support, Natural Language Processing (NLP), neural network chatbot, semantic similarity, supervisor–student matching.**

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1. Introduction

In higher education the element of supervision, especially in the proceeding of undergraduate and postgraduate research projects, is at the very core. Good supervision allows students to get a structured inquiry, form autonomous problem-solving ability and productively produce on their academic field. Nonetheless, assigning of students to supervisors in a manner that leads to fairness, efficiency and convergence interest is one of the great challenges encountered by universities all over the world. Historically, it has been a manual task to assign supervisors with references to the projects, i.e. the academic personnel or administrative staff manually goes over project proposals accepting them and tries to balance them with the available supervisor. The process is usually time consuming, subjective and constrained by what the managers have to know [1]. With the growth of a higher education institution and the number of students, these manual methods cannot offer fair measures, which results in a lack of correspondence between the experts and worsens the quality of the projects.

Besides the problem of allocation, communication among the supervisor and the students is also the key issue that determines academic success. Communications, feedback, and monitoring progress should be considered regularly to ensure that projects continue to move forward in terms of research [2]. Unluckily, scheduling, lack of time or misunderstanding can run counter to communication. Consequently, delays, confusion between students and supervisors and inefficiencies may occur and research activities cannot be carried out. Such concerns indicate that there is a necessity of a structured and technology-supported framework that may enable not only smart supervisor assignment but also an efficient communication process.

New technologies in processing texts and applications of computational linguistics hold the hope of finding solutions to these problems. The Natural Language Processing (NLP) has become a powerful approach of analyzing the unstructured textual data, so it is now feasible to extract some meaning, determine the themes and calculate the similarities among the text units [3]. NLP can be used in an academic supervision setting

to analyse the project descriptions of students and match them with supervisor interests thus facilitating automated, interest-based matching. This will change the process that is based on manual checking and will incorporate a more procedural and transparent system of allocation.

Also, introducing the structured communication tools to the supervision process has been highly suggested in educational study. The internet-based services that enable timely updates, feedback, and participation tracking were found to enhance the collaboration and lessen the hurdles of administration [4]. These platforms coupled with automated matching of supervisors have the potential of forming a comprehensive framework that not only makes sure that supervisor resources are allocated appropriately but also keeps guidance efficient throughout the entire process.

The offered system is aimed to address this gap by creating a web application on the basis of NLP-based supervisor assignment and an integrated communication channel of guidance communication. Students will be in a position to provide their project topics which will then be processed to provide the keywords, semantic meaning, and thematic relevancy. Supervisors will also be structured into their areas of expertise, past work they have done, and the time they have to offer. The two data streams will be compared by the system to give a list of ranked recommendations per student and the allocation can be done more feasibly and equitably.

Simultaneously, the creation of communication module will be included in the platform, which allows real-time guidance. This would not only aid communications between the students and the supervisors but can also come in handy by offering regular support that would answer the most common questions by generating quick responses, reminders as well as informing the project milestones. This way of approach reflects the duality of supervision: although at the beginning both expertise alignment is paramount, consistent communication is also vital in the course of the project.

The field of allocation of supervisors and guidance has gained more consideration in the recent years. Research has also stressed on the value of organized allocation

mechanisms to avert mismatch with possibilities to impact both the academic staff and student negatively [5]. In the same vein, digital guidance systems have been implemented in vocational and continuing medical education and corporate training with demonstrable increases in efficiency and satisfaction [6]. Implementing the provided ideas into the area of academic project supervision, the offered system solves a half-century old dilemma in the field of higher education.

The implementation of the given systems is also in line with the overall trends of digital transformation on university levels. The trend in the world is to shift institutions to online programs to conduct courses, administer and engage students [7]. The alignment and communication system between supervisors and students completes these developments, allowing administrators to be more on top of things and gaining the student more access to guidance. Notably, it also comes with a decrease in the amount of manual work that supervisors will be performing as they are able to devote themselves to offering expertise instead of administrative organization.

The given idea can serve the field in two ways: by addressing two key aspects: (1) automated supervisor assignment according to the textual representation of both the project topic and expertise of the supervisor, and (2) Project Guidance rolled into one with accumulation tracking of guidance contacts. As opposed to generic allocation tools, the proposed platform would focus on interest-based matches, where not only the number of available holdings to work with content items was aligned with the supervisor core topic but also when thematically relevant. Moreover, the use of communication tracking will be used to foster and support a relationship once an allocation has been done.

It is expected that the result will be a system that reduces transparency, impartiality, and efficiency in assigning supervisors, and at the same time enhances communications and guidance. The abilities of the students to access the appropriate expertise and prompt feedback are supposed to improve, and the supervisors are likely to see the decreased administration burdens and should have a more evident understanding of their

supervisor obligations. Universities will have a well-established setup to facilitate quality assurance, accountability, and better results of the projects.

In short, difficult issues of efficient supervisor assignment and guidance in the academic environment require new problem solutions, which combine the field experience with organized support mechanisms of communication. The presentation of the proposed platform directly solves these issues through the application of modern methods of text processing and design of communication systems. It provides a solution to institutions trying to manage increasing student population, shortage of supervision staff and the emerging need of transparent and fair academic procedures.

2. Background & Literature survey

2.1 Background

Guidance and supervision have always been cited as major aspects of academic achievement, especially those that concern higher education and research-based settings. In the past, such activity as assigning supervisors to students was performed manually, generally depending on the faculty availability, organizational policies, or overall subject alignment. Although such a strategy could be applied when cohorts of students were relatively small, it creates considerable difficulties with the recent boom in higher education launched in the last decades. Growing enrolment rates, increased diversification of areas of research enquiry as well as the need to provide supervision in a timely fashion have brought to the fore the inefficiencies of manual allocation techniques [2]. Mismatches between the supervisors and students in most of the cases led to a bad match in terms of expertise, lower motivation levels, and delays in the project.

In later 20th century, universities started to test semi-structured allocation methods, usually implementing faculty databases and search mechanisms based off keywords. Students and supervisors were clumsily paired within these systems using research interests and the spoken keywords. Nevertheless, even these keyword methods had shallow coverage because they did not have the larger semantic interpretation of project descriptions and the results could give false or low specificity matches [8]. Simultaneously, there was little communication between students and supervisors other than face-to-face meetings, letters and later emailing. Although email brought a certain level of flexibility, not every time was it effective in giving timely feedback or when communicating research projects due to the dynamic nature of communication.

The development of e-learning tools in the early 2000s was the beginning of digitalization of higher education. Moodle, Blackboard and Canvas were all tools that offered a structured method of communication, which however, were not constructed to meet the challenges of supervision and guidance. Such platforms did not usually have

personalized design or intelligent matching capabilities, so the supervisory task was left to the administration staff and faculty members to do most of it [9].

The developments in Natural Language Processing (NLP) in the last decade have given a more advanced tool with which to analyse academic text. The methods of semantic similarity, topic modelling, and transformer-based architectures allow interpreting the descriptions of projects performed by students with more accuracy, which results in the better distribution of supervisors [10]. Along those lines, conversational agents and chatbots have become useful in educational settings offering real-time assistance, guidance and interactive learning. Chatbots of a certain type, neural network-based to be precise are able to support guidance -related communication in real time due to the ability to support a context-aware conversation [11].

Generally, the shift in the methods of manually driven to the intelligent, AI-aided systems represents a wider tendency in academia of data-based decision-making and technological learning enhancement. The history of this project thus finds its origins in the lack of solutions to long held issues of supervisor and student alignment, poor communication, and lack of progress monitoring by circularly utilizing NLP, neural networks, and predictive analytical methods to provide a more cohesive and successful supervision setting.

2.2 Literature review

2.2.1 Domain Analysis

Over the past years, the study scenario in the area of supervision, guidance systems, and intelligent communication has widened, having contributions advanced by both areas of educational technology as well as computational intelligence. The three major studies are behind the scope overview in the present project as reviewed below.

Supervisor–Student Matching using NLP

Nguyen, Gardner, and Sheridan (2017) examined how to employ topic modelling to match research students to the possible supervisors on the basis of a thesis proposal and research propose of faculty member [12] s. In their research article, they used Latent Dirichlet Allocation (LDA) to induce themes of research in the student documents and contrasted them with the publication record of supervisors. The system proved to be more accurate as compared to manual allocation of funds which is evident in the large universities with a wide range of research fields. Nonetheless, the use of topic modelling restricted semantic interpretation because slight changes in a word choice would result in a misfit. This just reemphasizes the necessitation of a more sophisticated NLP formula, e.g. transformer-based models, which are more adept to identifying more profound relations of semantics. This is the start that the current project makes by implementing the NLP algorithms, which allow breaking down and conducting semantic similarity comparisons, thus improving the accuracy of supervisor distribution.

Guidance Communications Chatbots

Ranoliya, Raghuwanshi, and Singh, ran an extensive review of chatbot usage in education, where it is noted that chatbots can work well as providing immediate response, aiding in the process of learning, and ensuring uninterrupted communication [13]. According to their results, rule-based chatbots were not effective to process complicated scholarly conversation, whereas neural-network-based chatbots were more versatile and conscious in the context. Regardless of the promises, the majority of the

implementations were not integrated with guidance -specific needs like the discussion of projects, the exchange of feedback, and research guidance. It is indicative of the need of the domain-specific chatbots to be designed to support a supervisor-student interaction. This gap is proposed to be closed in the given project because the proposed guidance -oriented chatbot aims to use neural networks and NLP to establish meaningful academic conversations conditions in a real-time chat environment.

Neural Network–Driven Chatbots for Educational Guidance and Student Support

Recent implementations of neural-network chatbots in educational environments demonstrate considerable promise for scalable guidance. Neumann *et al.* developed FeedBot and LitBot to support over 700 students with self-study and written tasks, showing that automated agents can supplement limited human supervision with timely feedback and guidance [14]. Peyton’s review also highlights the expanding use of chatbots for student support in university settings, noting their capacity to evolve from administrative FAQ tools to more conversational, NLP-enhanced assistants [15]. A broader review by Davar *et al.* underscores both the pedagogical potential of real-time tutoring chatbots and the challenges of ethical deployment and data privacy within educational contexts [16]. Meanwhile, Aleedy provides insight into how chatbots assist educators by analyzing discussion threads and facilitating real-time interactions with students using NLP techniques [17].

2.2.2 Technical Review

This research formulates SSAP as a multi-objective binary constraint optimization problem to resolve the biases in regards to preference satisfaction, capacity restrictions, and illusion of workload. Manual allocation is pushed by a tunable trade-off parameter and a real-world decision-support tool reveals sensitivity analysis so that administrators can view how choices vary as priorities vary. The manner of formulation involves the introduction of constraints, including quotas based on the number of supervisors and quotas on the number of students and solution frontiers are explored to disclose near-Pareto decisions. Empirical analysis using a real data of postgraduates depicts clear

control of competing criteria. One of these is semantic alignment being that what is inputted is organized ratings/preferences vs. semantics of topic do not come out as proposals or publications. In application to research, this model would be well-matched with an upstream NLP similarity module to translate project textual project match to profile matches into quantitative scores which can be optimized [18].

The review is a systematic review of the empirical and design literature of education amid chatbots, which defines the technical patterns strategies in the classroom and advising setting. The results demonstrate that the chatbots have the capacity to provide on-demand help, guidance, and reduce the workloads of instructors. However, its drawbacks are that the accuracy of responses is not always equal, doubts are cast on the assessment fairness, and issues of privacy and ethics management arise. Technically, chatbot systems are scalable in terms of rule-based dialogue managers that extend to neural conversational agents that have context tracking: in applications where chatbots are used as mentors, the review notes that the domain-specific prompts used or relevant escalation paths as well as the audit logs are significant. In the case of a supervisor-student setup, these findings encourage the application of guardrails which include consent, rich logging, the existence of a failure path back to human experts, and data-minimization methods. Along with this, evidence base supports the necessity of metrics that are not measured by matching the intent, but ensuring the communication quality, user engagement and trust signals are observed [19].

Recent work shows that student-facing chatbots are both widely adopted in universities and capable of delivering measurable support, making them well-suited for a supervisor-student guidance channel. A large multi-institution survey by Stöhr, Ou, and Malmström reports substantial uptake of chatbots [20], across disciplines and study levels, with students valuing immediate responses for routine queries and course-related clarifications—evidence that conversational assistants can reliably absorb day-to-day load and free supervisors for higher-value guidance. Meanwhile, a randomized controlled trial (MISHA) demonstrates that a student-focused coaching chatbot can produce medium effect-size improvements in stress and depressive symptoms over

several weeks, indicating that structured conversational interventions can sustain engagement and deliver outcomes when human guidance bandwidth is limited. Together, these findings support a design in which the platform routes routine guidance questions to a neural network–driven chatbot, escalates edge cases to supervisors, and logs dialogues for transparency—achieving responsive communication without overburdening staff [21].

2.2.3 Similar Systems Study

1. Final-year project management systems (FYPMS)

The institutions commonly use FYPMS to aid in the process of supervisor assignment, registering of projects, and overall administration. Supervisor requests in the systems, the students will usually request supervisors by manual email requests and the administrative authority will make the assessment of acceptability based on workload ability or on keyword basis. Despite FYPMS being useful in terms of keeping records and process control, it fails to take advantage of advanced semantic analysis in order to facilitate desired match between the supervisory specialization and student subject interests. Comparatively, the proposed system will be able to include NLP-based matching, real-time communication processes [22].

2. EasyChair

The EasyChair academic conference management system, used extensively with academic conferences, includes uploading and reviewing of papers, allocation of reviewers and facilitated interaction. Its assignment process is mostly rules-based and bases the choice of reviewers (or, more aptly, supervisors) based upon availability and stated expertise. EasyChair has poor messaging and workflow capabilities but lacks semantic matching with NLP capabilities and Real-time communication with mentoring. In addition, predictive evaluation is not available, such as the analysis of sentiments. Thus, EasyChair does not focus on proactive guidance and other academic assistance but highlights administrative workflows [23].

3. Microsoft Teams (Version 2)

Microsoft Teams is a specifically adapted educational format with the ability to collaborate based on instant messages, video conference, and document sharing. Although it is so efficient in case of everyday usage, the platform does not include supervisor-student distribution option, and does not predict the performance through predictive models. Teams promotes effective interactions between the students and supervisors but it is still unable to match the projects with the supervisor exposures or score the progress with the such models. The suggested system aims at solving these gaps, as it combines real-time communication with academic-specific matching features and tracking opportunities [24].

4. Moodle + Chatbot.

The learning management system Moodle has also been enhanced with chatbot integrations within various institutions to help students navigate courses, offer reminder suggestions and answer frequently asked questions (FAQ) to save time. As much as these chatbots can supplement the level of accessibility and provide support in a timely manner, they were not specifically designed to support research guidance or to assist supervisor and student in collaborative work. Moreover, Moodle does not have the supervisor assignment based on natural language processing (NLP). Moodle is broad on the educational requirements compared to the proposed system which fails to target the supervision and project assessment challenges which require special consideration [25].

5. Brightspace by Desire2Learn (D2L)

Brightspace is a modern online educational platform used in the higher education system around the world. It is the aggregate of communicative, feedback, and analytic functionalities to aid instructional practice. Although Brightspace also uses learning analytics to track engagement and performance of the students, it does not pay that much attention to supervision of projects as it is centered at courses. It lacks NLP-based supervisor assignment and commitment guidance chatbots. The offered solution is

different since this is a targeted solution that can be used in project-based guidance, which includes matching, communication, and predictive monitoring [26].

Comparative Analysis

Existing platforms each contribute partial solutions to supervision and communication in academic contexts, but none fully address the dual challenge of accurate supervisor–student matching and continuous guidance support. Final-Year Project Management Systems (FYPMS) enable record-keeping and allocation through keyword filters or administrative judgment; however, they lack semantic depth and integrated communication. EasyChair provides rules-based reviewer allocation but offers minimal mentoring functionality, making it more suited to administrative workflows than ongoing supervision. Microsoft Teams facilitates real-time communication effectively, yet it is a generic collaboration platform with no mechanisms for supervisor allocation or project-focused guidance. Moodle, with chatbot extensions, supports FAQs and reminders, but these bots are course-oriented and not tailored for research guidance. Similarly, Brightspace emphasizes engagement analytics for coursework but overlooks project-based supervision needs.

In contrast, the proposed Supervisor–Student Matching and NLP-Based Project Guidance System uniquely integrates semantic matching through NLP with a neural-network-driven chatbot for real-time academic guidance. Unlike existing tools, it combines automatic supervisor allocation based on expertise and availability with continuous conversational support targeted at research supervision. This dual focus on matching and guidance makes the proposed solution stand out, directly addressing the gaps in current systems by ensuring that project alignment, supervisor expertise, and student communication are seamlessly integrated into a single platform.

The proposed system uniquely combines these capabilities into a single academic supervision platform.

2.3 Research Gap

System / Feature	Supervisor Matching	NLP-Based Matching	Real-Time Communication	Chatbot Guidance	Project - Specific Focus
Final-Year Project Management Systems	✓ (manual/keyword)	✗	✗ (email only)	✗	✓
EasyChair	✓ (rules-based)	✗	✗ (basic messaging)	✗	✗
Microsoft Teams (V2)	✗	✗	✓ (chat/video)	✗	✗
Moodle + Chatbot	✗	✗	✓ (with chatbot add-on)	✓ (basic FAQ bots)	✗
Brightspace (D2L)	✗	✗	✓	✗	✗
Proposed System	✓	✓	✓	✓	✓

2.4 Research Problem

Supervision is a critical parameter in establishing the effectiveness of an academic project but most institutions still use manual or half automated systems to assign supervisors. Such administrative practices based on administrative judgment, on searches based on keywords or on equal distribution of work are not adequate to reflect the semantic connection of a research topic with a research supervisor or supervisor expertise. Therefore, it ends up with mismatches, which create misaligned steering, lack of efficiency, and poor project results.

At the same time, the communication between the students and supervisors is fragmented, distributed in emails, separate meetings, and outside platforms that do not have any functions developed specifically to help guidance. With this kind of an arrangement, there is a delayed feedback, inconsistent engagement levels, and there are no moderations of support united. Although project-management systems, collaboration tools and learning-management systems partially covered some of these issues, they were deficient because they could only provide partial solutions that could not integrate intelligent matching with continuous guidance into a comprehensive system.

The resulting research issue is that there is a lack of a specific system to combine the NLP-based supervisor matching and a chatbot based on a neural network to have the possibility of real-time communication and guidance. Without it, institutions continue to be inefficient in the process of allocation of supervision and in other aspects of student engagements to foster a gap that compromises the achievements made in academic projects development.

2.5 Objectives

2.5.1 Main Objectives

To design and develop a supervisor–Student Matching and NLP-Based Project Guidance System that enhances academic project supervision through intelligent supervisor allocation and automated topic-support assistance.

2.5.2 Specific Objectives

- To implement an NLP-based algorithm that semantically matches student project topics with supervisors according to expertise and availability.
- To develop a neural-network-driven chatbot capable of answering student queries and clarifying doubts related to project topics and sections.
- To establish an interest-based ranking system that improves the alignment between student research needs and supervisor specialization.
- To integrate supervisor allocation with a student-facing support chatbot in a unified platform that streamlines supervision processes.
- To evaluate the effectiveness of the system in improving supervisor–student alignment and in providing timely academic assistance through automated responses.

3. Methodology

3.1 System Overview

It is suggested that a refined mechanism of matching supervisors-to-students and applying NLP-based Project Guidance system should be offered, where specific academic support can be provided to a student through the mentoring chatbot based on the matching technology. The two modules integrated to comprise the system are a Natural Language Processing (NLP) engine to dispatch supervisors and a chatbot based on a neural network to answer topic-related queries.

On enrolment, project titles and abstracts are handed in by the students and processed by the NLP module that extracts and codifies the semantic information in it. This is then cross-check with a database of supervisors' areas of specialty and current workload situation. A list of the most-appropriate supervisors is created that will be made, subject to the semantic applicability of allocations, in such a way that it is able to balance allocation of the resources along the institution.

Afterwards, the students are given access to an automated chatbot interface after allocation. The chatbot uses a model of a neural network and answers questions related to the research processes, project design, and the rules of academic writing. The support can be in terms of clarifying the descriptions of methods, organization of literature reviews and alignment with technical standards of formatting. Automatizing non-unique academic consultations, the chat bot will free the supervisors of the repetitive administrative work and allow applying efforts on top-value mentoring.

This duality in architecture of the system, the germane supervisor guidance and bot-mediated support that is scalable, facilitates on the one hand operational efficiency and on the other hand prevents mismatched supervisions and increases the overall experience of guidance.

3.1.1 System diagram

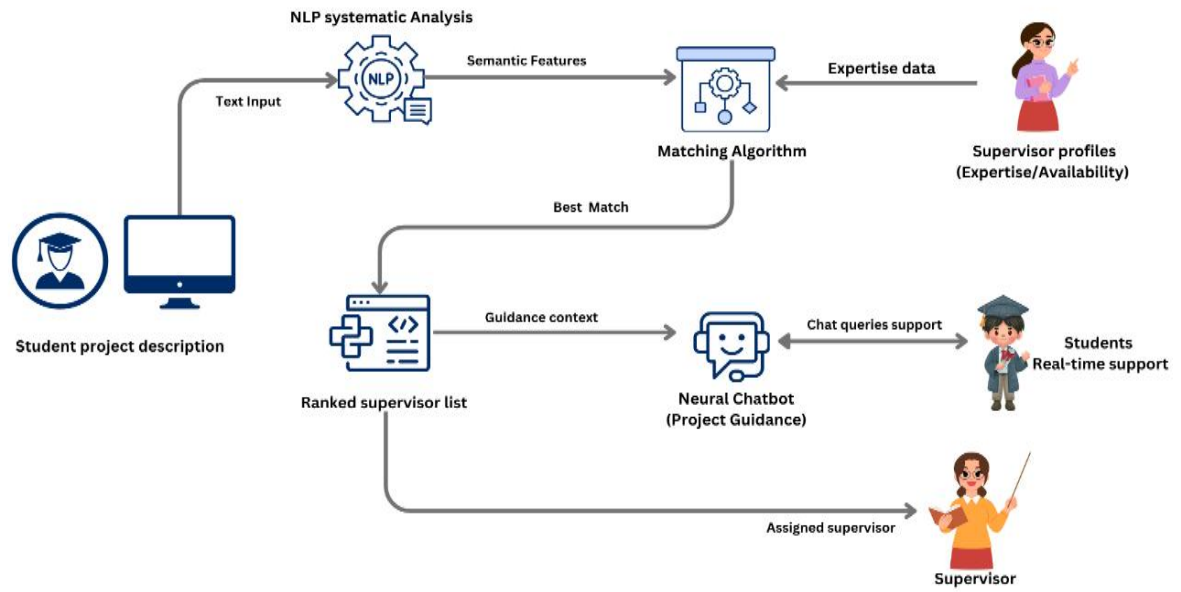


Figure 1: System diagram

3.1.2 System Architecture

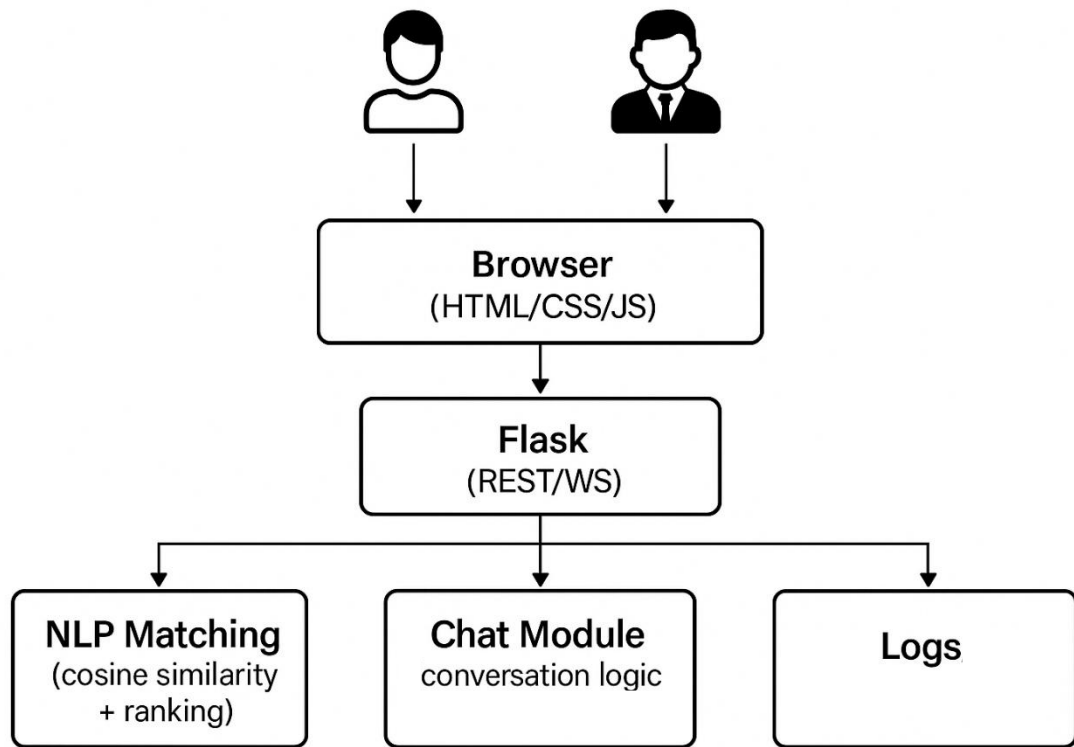


Figure 2: System Architecture

3.2 Technologies

3.2.1 Frontend Development

The system frontend, which will be developed using HTML, CSS, and JavaScript, will present a user-friendly feature, through responsive design to individuals accessing it, be it students or supervisors. This layer forms the main interaction point and it is accessible and convenient to use in any devices. The students will be in a position of posting their project topics using structured forms and the supervisors will manage and update their expertise profiles to be accurately matched. It will have a specialized chat interface, which will allow communication in real-time guidance, thus establishing a smooth environment of clarifications, feedback, and guidance. Besides, a performance dashboard will be included in frontend to show the forecasts and warnings in the backend modules, as well as to present visual advertising on the project progress and potential threats. The frontend will facilitate the usability, interactivity, and visualization where the system will be practical to maintain continuity in the work between supervisors and students.

3.2.2 Backend Development

The backend will be created in Python Flask that would allow a lightweight and flexible REST-based frame to handle the application logic and integration with the analytical modules. This element will play the role of the central processing unit in interlinking frontend with the matching engine, communication module and forecasting functions. It will be receiving proposed topics of project and preprocessing the input text and sending it to be checked with the similarity. Communication between the chatbot and users will also be controlled by the backend because it will be in charge of processing the messages sent and received in real-time.

3.2.3 NLP for Supervisor–Student Matching

Natural Language Processing (NLP) Pipeline

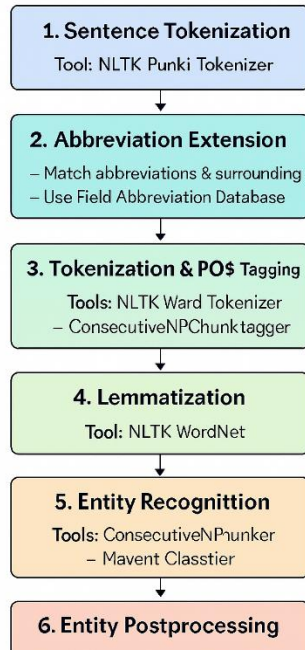


Figure 3: NLP for Supervisor–Student Matching

Project titles and descriptions submitted by students will first be cleaned using standard text preprocessing methods. This will include tokenization (splitting text into words), removal of stopwords (commonly used words with little meaning, such as *and* or *the*), and lemmatization (converting words to their base form). These steps will ensure that the input text is standardized for analysis.

Once the text is cleaned, it will be converted into numerical representations known as embeddings. These embeddings will capture the meaning of words and sentences in a mathematical form. Unlike keyword-based methods, this approach will allow the system to understand context — for example, recognizing that “machine learning” and “data-driven models” are conceptually related.

The embeddings of student topics will then be compared with the embeddings of supervisor expertise profiles using similarity measures such as cosine similarity. The

system will produce a ranked list of supervisors based on how closely their expertise aligns with the student's project description. This ranked output will guide allocation, ensuring fair and accurate matching.

The core of this system revolves around Natural Language Processing (NLP), which enables the machine to understand, interpret, and analyze human language. By using advanced semantic analysis techniques, the system evaluates the content of both project topics proposed by students and the research areas listed in supervisor profiles. This helps in finding the most suitable matches based on the textual similarities and relevance of the topics. To achieve this, preprocessing techniques such as tokenization (splitting text into smaller components like words or phrases) and embeddings (converting text into numerical vectors) are employed. These steps ensure that the system can process and understand the nuances of natural language. The latest technologies like BERT (Bidirectional Encoder Representations from Transformers) and Sentence-BERT are then applied to enhance the model's ability to understand complex sentence structures and contextual meanings, improving the accuracy of student-supervisor matching.

3.2.4 Artificial Intelligence & Deep Learning

The Artificial Intelligence (AI) and Deep Learning components of the system drive the decision-making process, ensuring the correct supervisor-student matches. Using deep learning algorithms, the system analyzes patterns and relationships between student projects and supervisors' expertise. The use of neural networks allows for continuous learning and improvement in the matching process. The system's ability to offer real-time guidance through an intelligent chatbot is powered by neural networks. The chatbot interacts with students, providing instant advice, resources, and suggestions for their project development. This real-time interaction enhances the student's learning experience by addressing questions as they arise, promoting a dynamic and supportive project environment. Furthermore, the implementation of advanced deep learning models like Transformers, RNNs (Recurrent Neural Networks), and RandomForest techniques ensures that the system can process large volumes of data and make accurate predictions about the best supervisor-student pairings.

3.2.5 Chatbot for Guidance Communication

The communication module will feature a chatbot to provide real-time interaction between students and supervisors. This chatbot will act as a first-level support tool and will operate in the following manner:

Conversation handling. The chatbot will process incoming student queries, generate context-aware responses, and provide clarification on supervision-related tasks such as submission deadlines or project milestones.

Training for academic context. To keep the responses relevant, the chatbot will be trained on academic and supervision-related datasets. This ensures that the conversation remains focused on guidance, support, and project management rather than general queries.

Integration with the system. The chatbot will be connected to the backend via REST APIs, enabling it to interact seamlessly with other modules. Through the frontend chat interface, students and supervisors will be able to access the chatbot directly, making communication faster and more efficient.

Fallback mechanisms. If the chatbot encounters queries it cannot resolve, the system will automatically redirect the conversation to the supervisor. This ensures that the tool enhances communication without replacing the critical role of human guidance.

3.3 Anticipated Conclusion

This recommendation proposes an institutionalized system that is meant to regulate power of supervisor relationships with the students. The system aims to address such problems as inappropriate supervision, reactive responsiveness, and a lack of an organizational communication channel by combining project-based matching procedures with constant real-time contact through a single web-based environment. The expected outcome is the more streamlined process as students will be guided by the supervisors, whose experience aligns with the goals of the project, as well as supervisors

will be provided with the tools that will help them communicate clearer and have better control of the process.

3.3.1 Results

According to preliminary literature, it is expected that the given system will help increase the accuracy of supervisor assignments and lead to more effective interaction between students and their supervisors. Since the matching of supervisors with students is based on the relevance of the project topic, there are reduced chances of mismatching thus ensuring that the students obtain meaningful and specific guidance. The integration of a chatbot is likely to speed up the responsiveness because the routine project-related questions will be managed by a chatbot, thus reducing the time waste and allowing the supervisors to focus on more meaningful academic assistance. As a result, there will be more general satisfaction of students and supervisors, as well as faster development of the projects.

3.3.2 Application

The system is most applicable in academic environments where large numbers of students require supervision and guidance. Universities and higher education institutions can benefit from improved workload distribution among supervisors and a centralized communication platform. The chatbot will further assist by handling common inquiries, allowing supervisors to concentrate on essential guidance tasks. This ensures smoother supervisory processes, reduces administrative overhead, and helps maintain consistent quality in project outcomes.

3.3.3 Real World Use

In real-life academic situations, the system in question is a feasible intervention that will aid institutions of higher learning (uni, college, training institutes), in the management of supervisor assignments and communications. It is flexible and can be applied to a vast range of disciplines and types of projects, which makes it suitable to a great number of academic programs altogether. Moreover, the similar structures can be applied to the realm of professional mentoring and training where the need to match the expertise with

mentee demands along with the need to establish smooth communications is also of utmost importance.

3.4 Commercialization

The suggested framework will be a formidable system designed in alerting and providing access to business opportunities. Its value is not limited to the scholarly environment; it may be sold to the government agencies and enterprises as well as nongovernmental organization who want to have strict governance systems and explore the management problems systematically. Learning institutions can use the platform on a subscription basis to facilitate the deployment of supervisors and education of students reducing administration costs and increasing the effectiveness of education. Its capabilities can also be applied in parallel, in professional-training organizations, in corporate internship programs, and research institutes, where mentors and progress management is essential. Commercialization strategies focus on scalable modes of deployment, such a cloud-based hosting, institutional licensing, and pre-packaging according to the requirements of particular industries. Estimates of the revenue generation involve subscription pricing structures, cumulative subscription, institutional licensing, and integration services. Figuring the system as flexible and application focused also changes the original research prototype into a commercially viable product whose market appeal is destined to be long-term.

4. Gantt chart

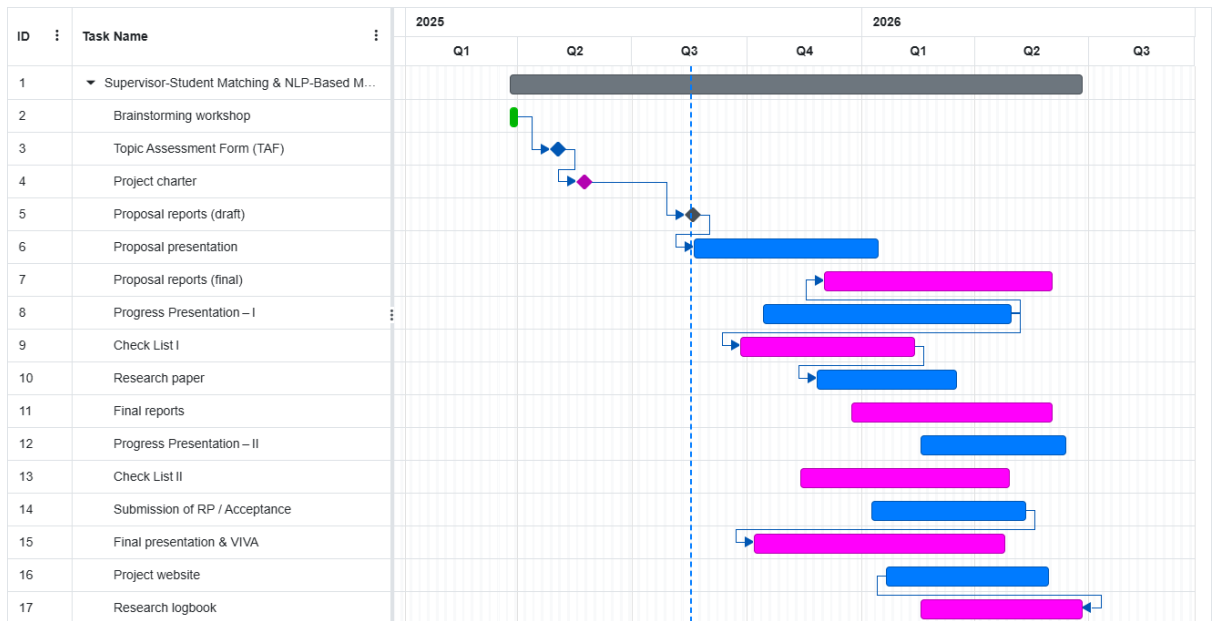


Figure 4:Gantt chart

5. Work Breakdown System

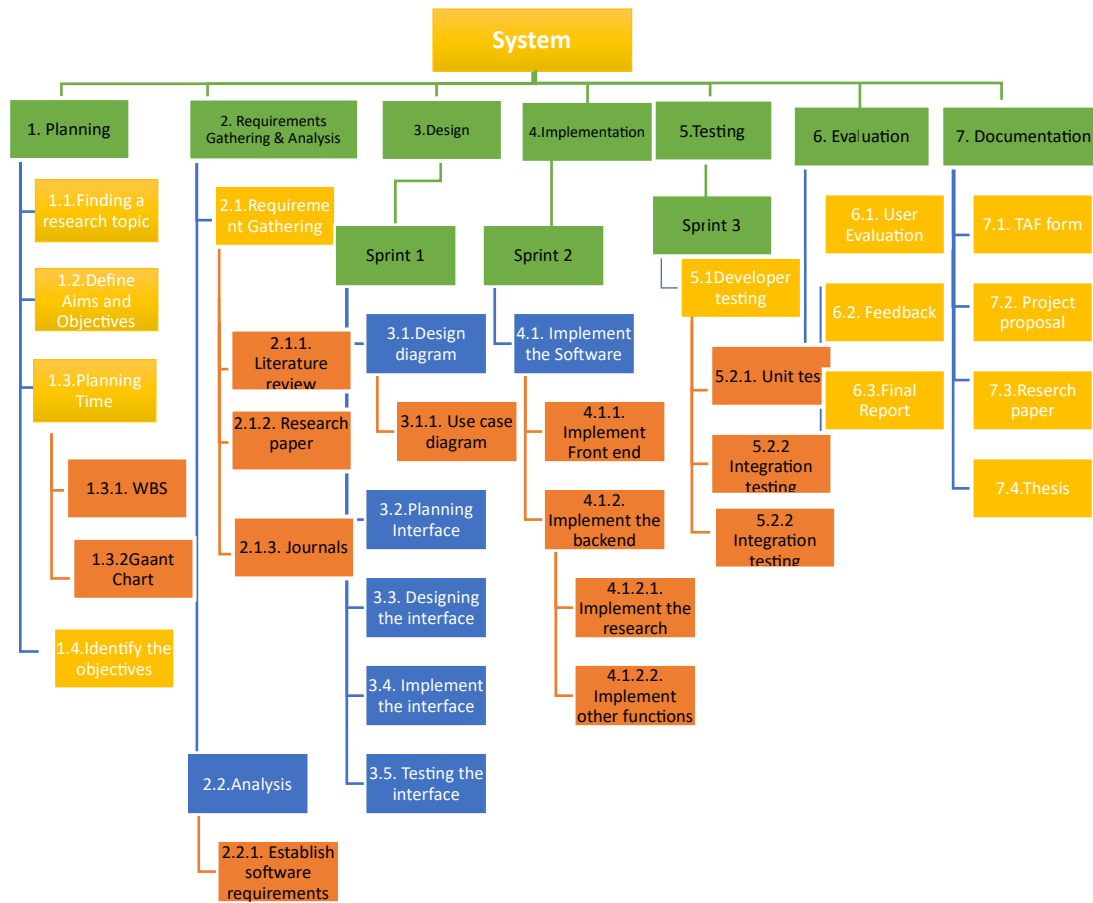


Figure 5: WBS

6. Project Requirements

6.1 Functional Requirements

Functional requirements define the core operations of the system that must be implemented to achieve the project objectives. The following table lists the main functions relevant to supervisor–student matching and communication.

Table 1: Functional Requirements

Requirement ID	Requirement Description	Priority
FR1	The system will allow students to submit project titles and descriptions.	High
FR2	The system will enable supervisors to register and update expertise profiles.	High
FR3	The system will process project descriptions and generate supervisor matches using NLP.	High
FR4	A ranking mechanism will be provided to suggest the most suitable supervisors.	High
FR5	The system will support real-time communication through an integrated chat interface.	High
FR6	The chatbot will handle common project-related and academic queries using neural network–based NLP.	Medium
FR7	The system will store communication logs for future reference.	Medium
FR8	A dashboard will display matched supervisors and communication history.	High

6.2 User Requirements

User requirements outline what different categories of users expect from the system. These requirements capture the needs of students, supervisors, and administrators in terms of access, interaction, and usability. By focusing on user expectations, the system will ensure a practical and effective solution that supports the end users' workflow. The table below presents the user requirements for the proposed system.

Table 2: User Requirements

User Requirement ID	Requirement Description	User Type
UR1	Students will be able to submit project topics and receive matched supervisors.	Student
UR2	Students will interact with supervisors via an integrated chat interface.	Student
UR3	Supervisors will be able to create and update their expertise profiles.	Supervisor
UR4	Supervisors will be notified of new student matches.	Supervisor
UR5	Supervisors will be able to respond to student queries through the chat interface.	Supervisor
UR6	Administrators will monitor supervisor allocation and system usage.	Administrator
UR7	All users will log in securely with unique credentials.	Student
UR8	Students will be able to use the chatbot for common doubts related to project work.	Student

UR9	System will ensure responsive access across devices (desktop)	All Users
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6.3 System Requirements

System requirements define the technical specifications and operational conditions under which the platform will function. These requirements cover software frameworks, hardware needs, security considerations, and performance expectations that ensure the system runs efficiently and reliably. The table below lists the system requirements for the proposed platform.

Table 3: System Requirements

System Requirement ID	Requirement Description	Category
SR1	The frontend will be developed using HTML, CSS, and JavaScript for an interactive user interface.	Software
SR2	The backend will be implemented using Python Flask with REST architecture.	Software
SR3	NLP techniques will process student project descriptions and match them with supervisor expertise.	Software
SR4	A neural network-based chatbot will provide automated responses for common project-related queries.	Software
SR5	The system will operate smoothly on modern web browsers (Chrome, Edge, Firefox).	Hardware /Client
SR6	The system will be hosted on a server with a minimum of 8GB RAM and multi-core processor.	Hardware /Server
SR7	All communications between students, supervisors, and chatbot will be secured using HTTPS protocols.	Security

SR8	The system will log supervisor allocations and communication records for transparency.	Operational
SR9	The platform will support multiple simultaneous supervisor–student pairs without performance degradation.	Performance
SR10	The interface will be optimized for both desktop and mobile accessibility.	Usability

6.4 Budget and Budget Justification

Table 4: Budget and Budget Justification

Item/Resource	Estimated Cost (LKR/USD)	Justification
Development Tools (IDE, Libraries)	25,000 LKR / 80 USD	Required for coding, debugging, and maintaining the web application. Includes access to licensed frameworks and essential plugins.
Hosting & Deployment (Cloud/Server)	45,000 LKR / 150 USD	Necessary for hosting the Flask backend, frontend, and machine learning modules. Ensures scalability and 24/7 availability during testing and demonstration.
Hardware (Laptop/Workstation)	150,000 LKR / 500 USD	A capable machine is needed to handle model training, running Flask services, and integration testing.
Internet & Communication	15,000 LKR / 50 USD	Supports collaboration, supervisor communication, and online deployment access.
Documentation & Reporting	10,000 LKR / 35 USD	Printing, binding, and preparation of final project reports and supporting documents.
Contingency (10%)	24,500 LKR / 85 USD	Allocated for unforeseen expenses such as additional hosting requirements, software updates, or minor hardware replacements.

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Appendix

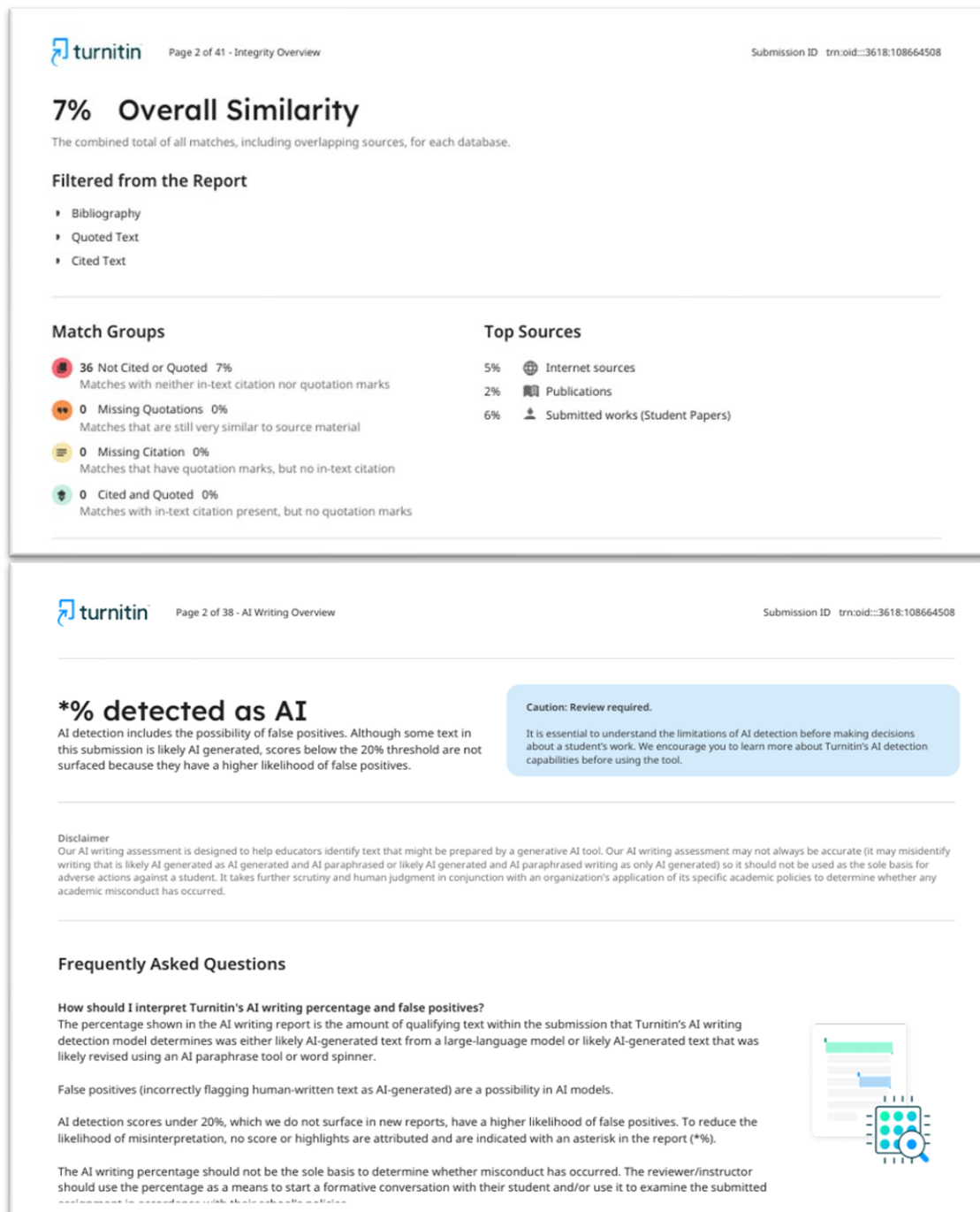


Figure 6: Plagiarism Report